We Need to Talk - Beta

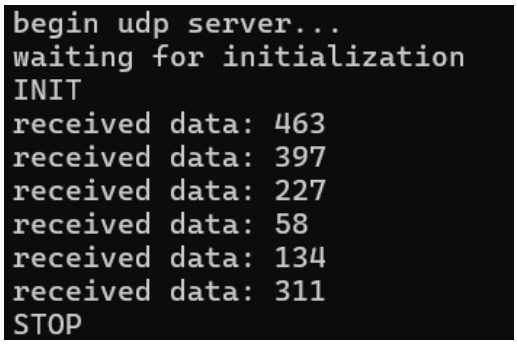
Autumn Rogers

Progress Summary

The current state of this project is:

- I have bidirectional communication between the Pi and the ESP
- The ESP is successfully reading and processing the photoresistor data (sending the average of the last five seconds every two seconds)
- The sensor readings and communication can be started and stopped with the use of a button on the Pi

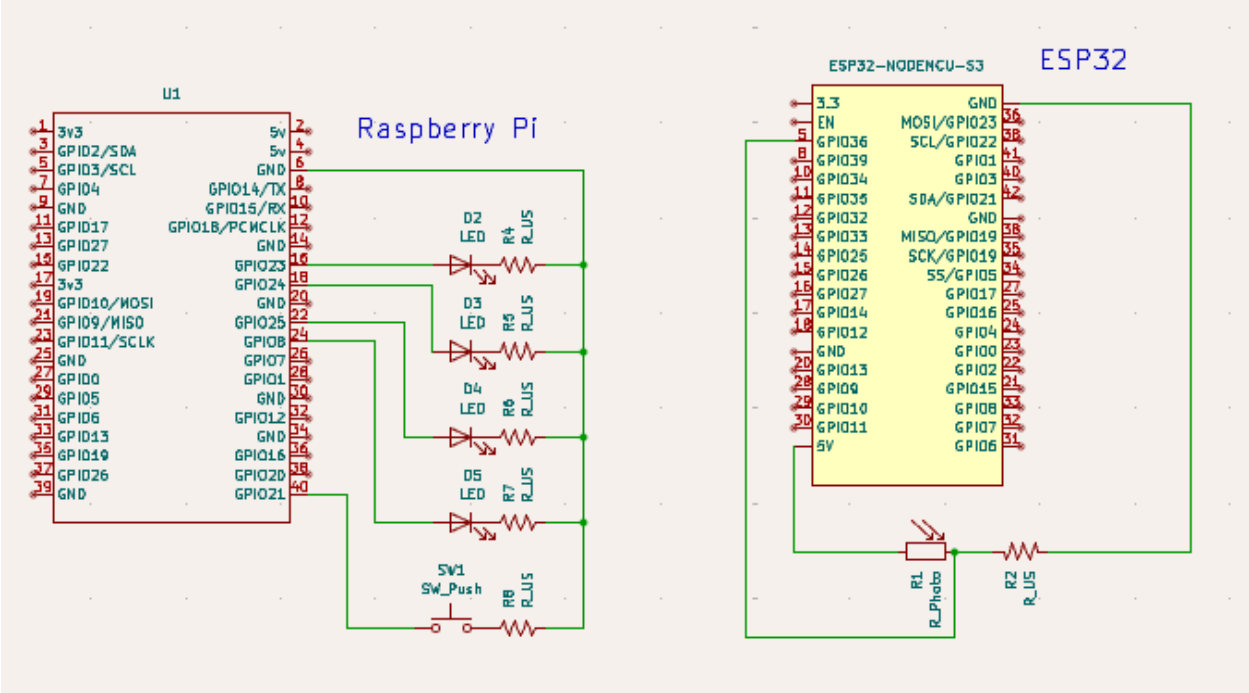
See the below screenshots of the Pi output (left) and ESP output (right)





Overview

Schematic

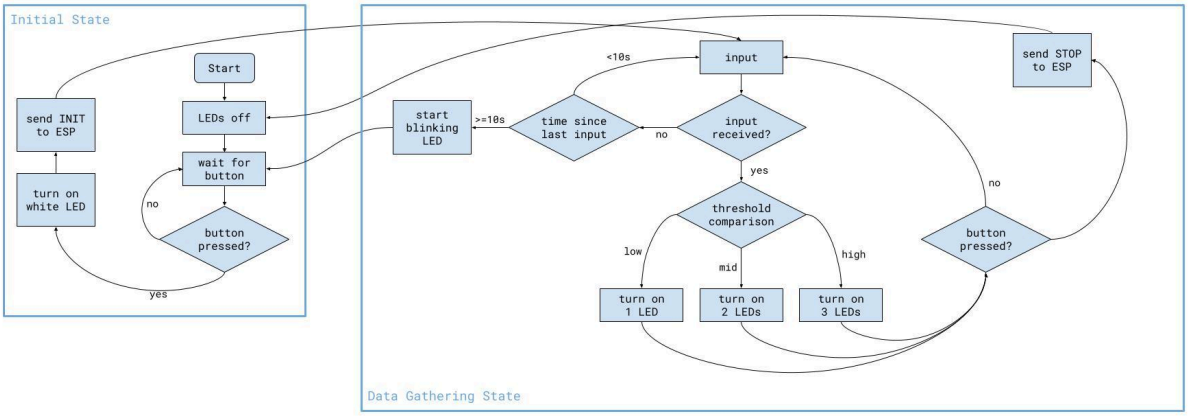


Communication Protocol

Sender	Receiver	Message	Purpose
Pi	ESP	`INIT`	Begin collecting data
Pi	ESP	`STOP`	Stop collecting data, return to ready state
ESP	Pi	`XXXX`	Integer representing the rounded average of the photoresistor data

Raspberry Pi WiFi setup and packet delivery

Flowchart



[Link to flowchart]

Initial State

In the initial state, the Raspberry Pi turns off all LEDs and waits for the button press, which will initiate communication by sending an `INIT` message to the ESP.

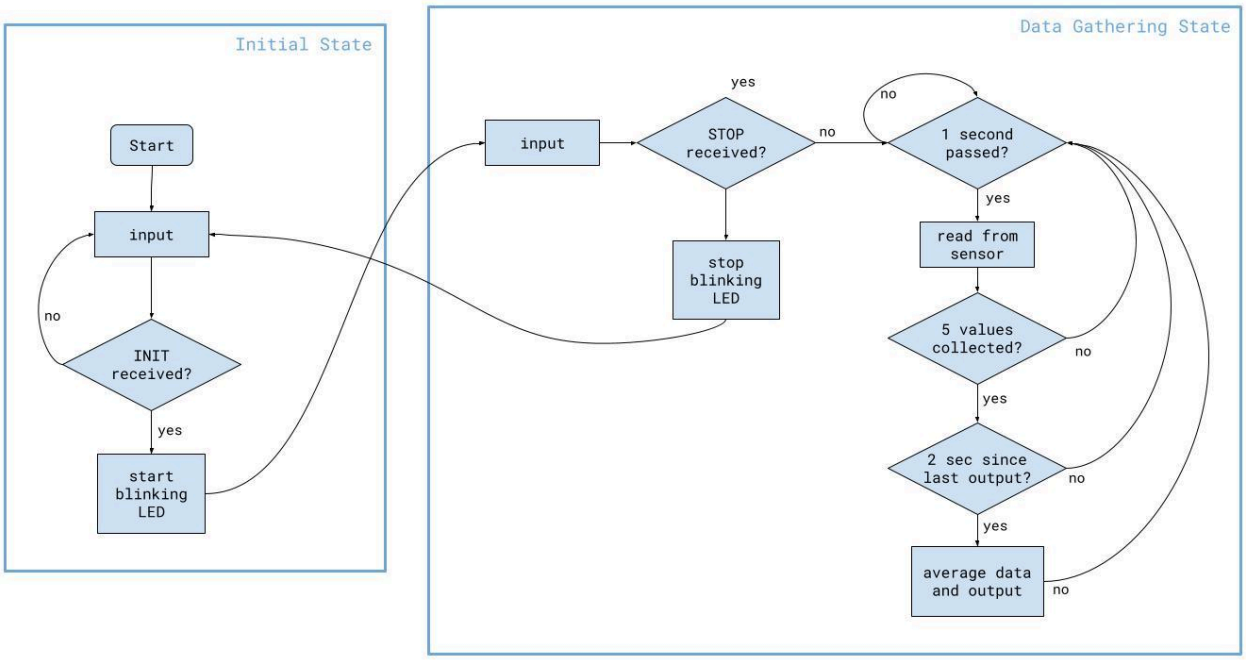
Data Gathering State

In the data gathering state, the Raspberry Pi listens for communication from the ESP.

- If it receives communication, it turns on 1-3 LEDs to indicate the brightness sensed by the ESP.
- If it doesn't receive a message for at least ten seconds, it begins blinking an LED to signify an error, and returns to the initial state until the button is pressed.
- While in the data gathering state, if the button is pressed again, it sends a `STOP` message to the ESP, and returns to the initial state.

ESP WiFi setup and packet delivery

Flowchart



[Link to flowchart]

Initial State

In the initial state, the ESP connects to WiFi and waits to receive communication from the RPi, indicated by an `INIT` message.

Data Gathering State

In the data gathering state, once a second, the ESP takes the current reading of the photoresistor. It stores the five most recent sensor values.

- Once at least five sensor readings have been taken, the ESP begins to output the average of the five most recent readings to the RPi once every two seconds.
- While in the data gathering state, if the ESP receives a `STOP` message from the Pi, it ceases data collection and returns to the initial state.